

FORMULATION AND STUDY OF A FROZEN DESSERT FROM HEMP MILK

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OBJECTIVES



Product Evaluation

Assess sensory quality and physicochemical stability of the ice cream

Functional Feasibility

Explore hemp milk's viability in frozen dessert systems and address formulation challenges

Optimized Formulation

Develop a hemp milk-based frozen dessert with standardized composition

INTRODUCTION

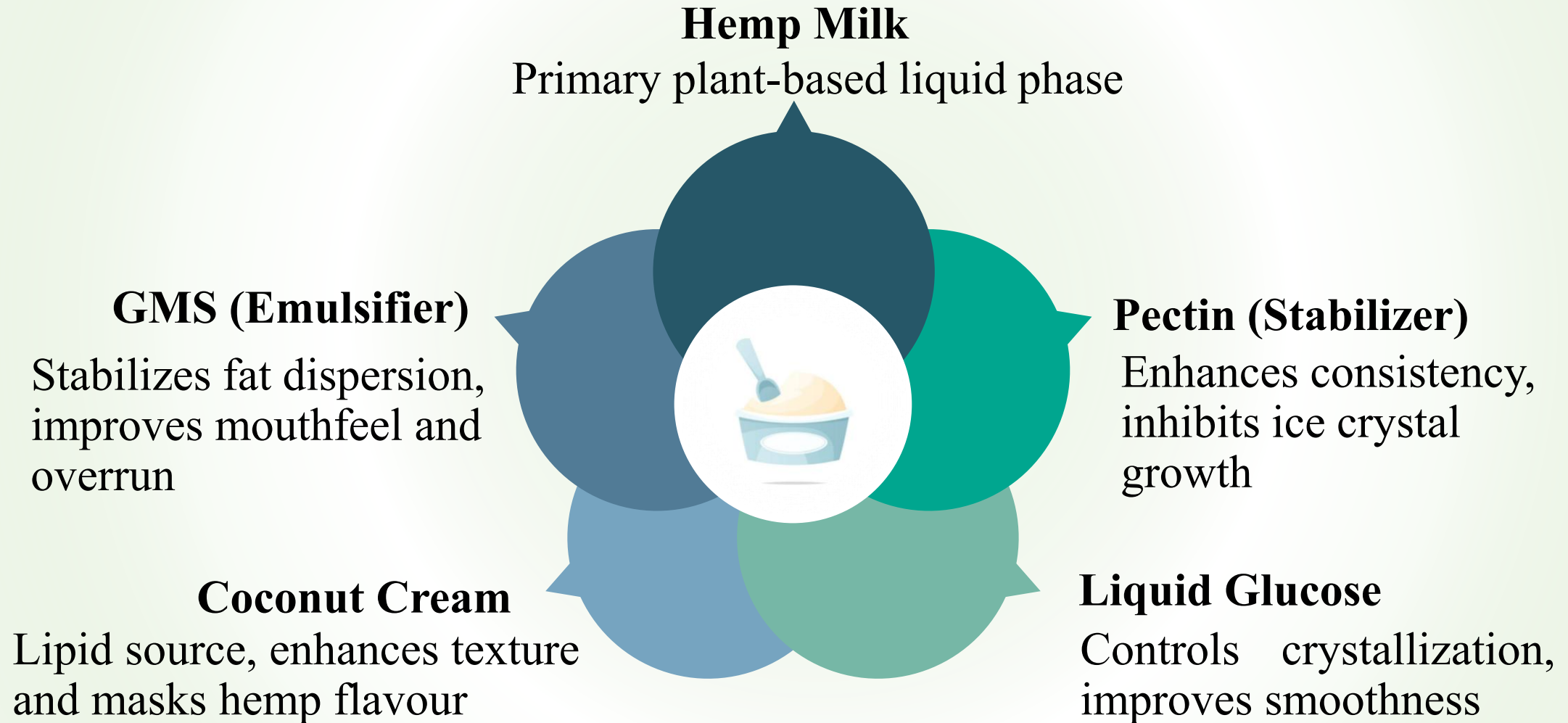
- **Botanical Origin:** Nutrient-dense seeds derived from *Cannabis sativa L.*
- **High-Quality Protein:** Contains all nine essential amino acids; classified as a complete plant protein
- **Essential Fatty Acids:** Rich in polyunsaturated fats with an optimal omega-6 to omega-3 ratio
- **Micronutrient Profile:** Significant levels of vitamin E, magnesium, potassium, and iron



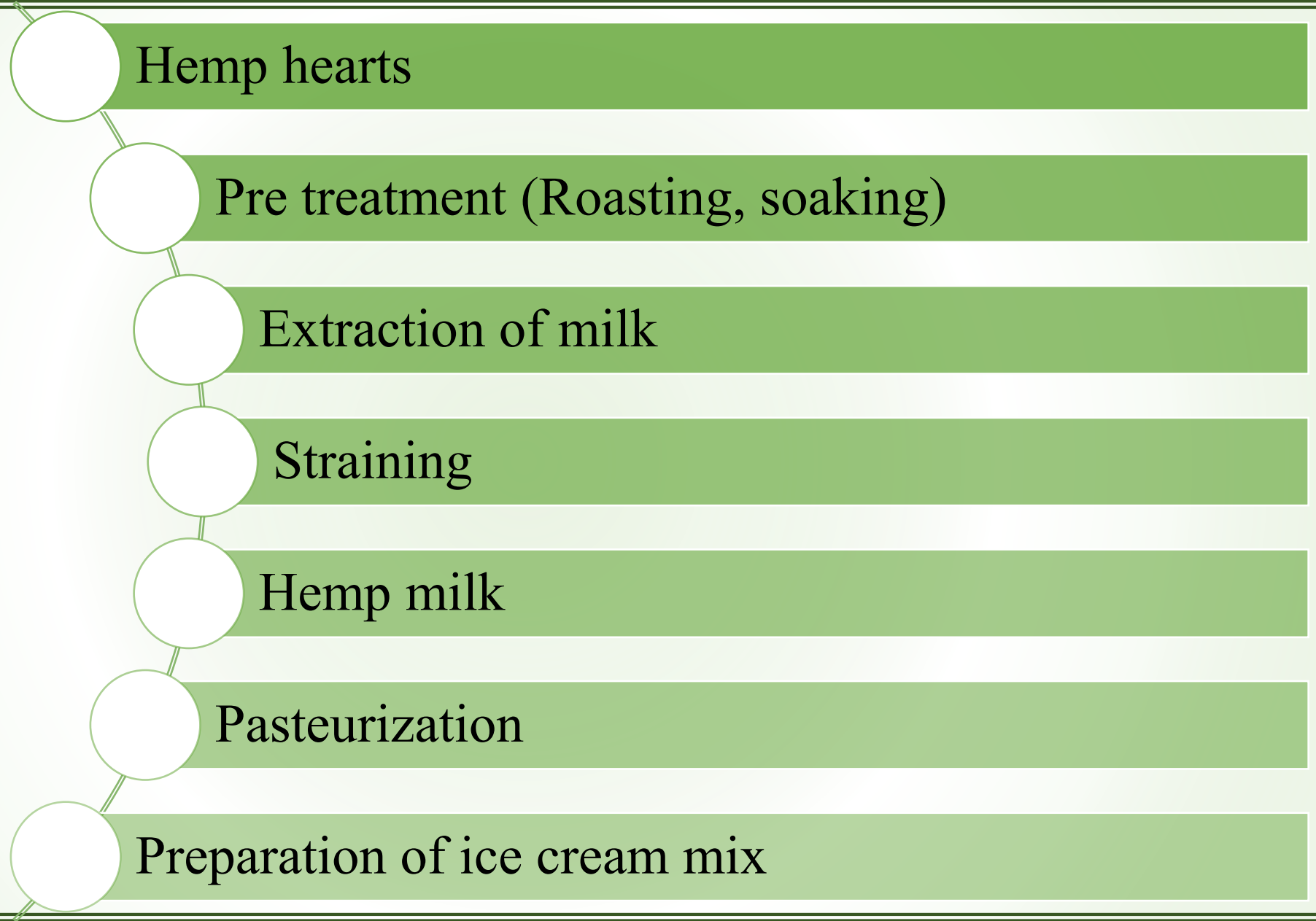
HEMP MILK

NUTRIENTS	HEMP MILK	COW'S MILK
Carbohydrates (%)	0.3–30.21	3.20–5.40
Protein (%)	0.83–6.96	2.9–3.32
Fiber (%)	0–0.4	0
Fat (%)	1.25–18.02	3.40–6.40
Ash (%)	0.47	0.67–1.9
Energy (Kcal)	19–54	63–118
Total Sugars (%)	2.5	3.4

INGREDIENTS



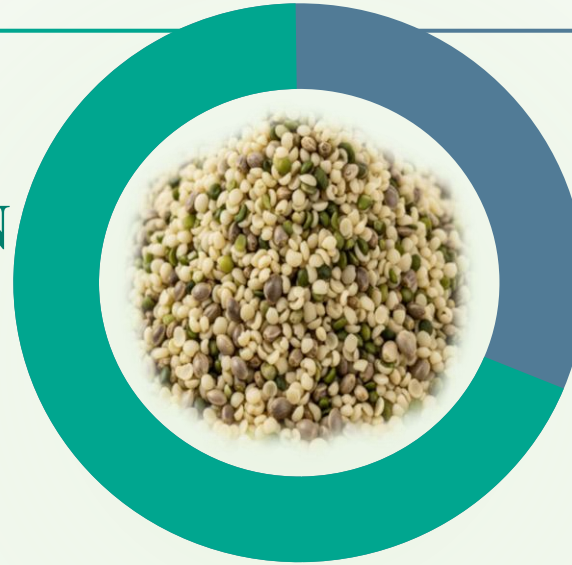
FLOWCHART



EXPERIMENTAL DESIGN & OPTIMISATION STRATEGY

ITERATIVE FORMULATION TRIALS

- 8 variants with systematic changes in sweeteners, fat %, stabilisers & emulsifiers
- Focus on balancing functionality and sensory appeal in plant-based matrix



SENSORY-LED OPTIMISATION

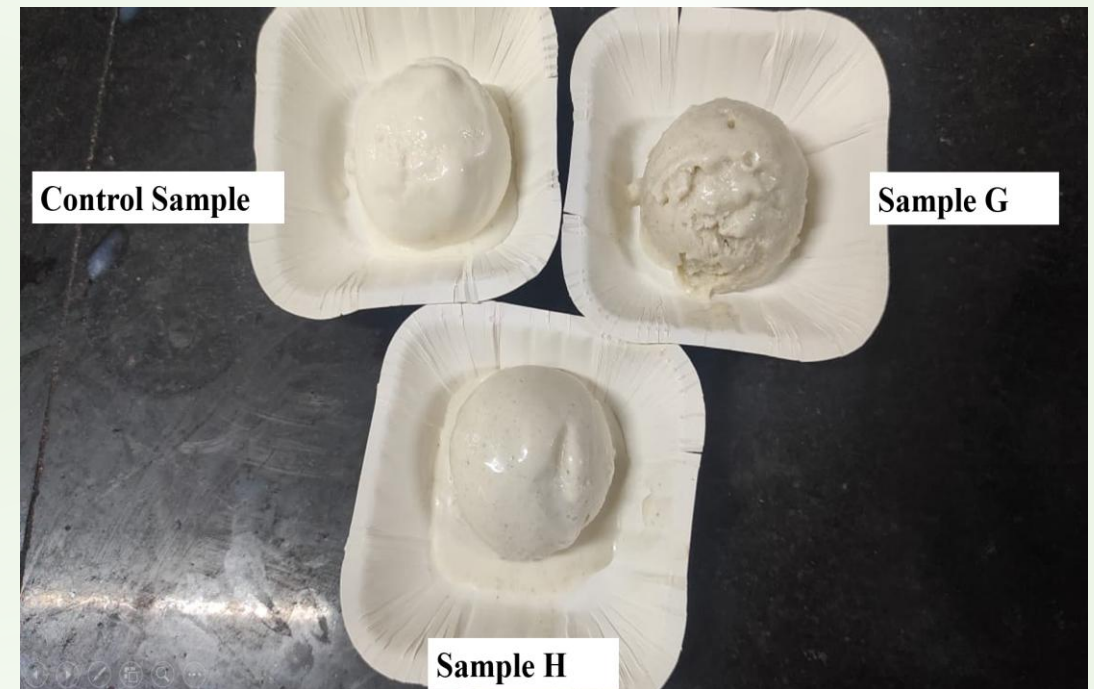
- Comprehensive evaluation: odour, flavour, texture, appearance, aftertaste, stability
- Progressive refinement to address flavour masking and texture enhancement challenges

SAMPLES

	A	B	C	D	E	F	G	H
Sweetener	Sugar 100g	Sugar 100g	Sugar 100g	Sugar 100g	Sugar 100g Liquid glucose- 15ml	Sugar 100g Liquid glucose- 15ml	Sugar 80g Liquid glucose- 20ml	Sugar 80g Liquid glucose- 20ml
Stabilizer/ Emulsifier	Stabilizer mix 1.25 g	Pectin 20 g	GMS 1.25 g	GMS- 1.25g Pectin- 1.25g	GMS- 0.75g Pectin – 1.25g	Pectin- 2.5g	Pectin- 2.5g	Pectin- 0.75g GMS- 1.25g
Coconut milk/ Coconut cream	Coconut milk 100ml	Coconut milk 100ml	50 ml (10% fat)	50 ml (10% fat)	100 ml (10% fat)	100 ml (10% fat)	125 ml (10% fat)	125 ml (10% fat)

KEY FINDINGS – PROCESS OPTIMISATION

- **Raw Material:** Hemp hearts preferred to eliminate hull-induced bitterness and discoloration
- **Pre-treatment:** Roasting (7 min, medium heat) yielded optimal flavour and nutty aroma
- **Extraction Ratio:** 1:10 (hemp:water) produced smooth texture and minimal aftertaste
- **Pasteurisation:** 72°C for 15–18 seconds ensured microbial safety without coagulation



KEY FINDINGS – SENSORY OPTIMISATION

Early Formulation Issues (A–B)

- Hay-like odour, beany off-notes, grainy texture, and phase separation

Mid-Stage Refinements (C–F)

- Liquid glucose + coconut cream improved flavour, masked aftertaste
- Pectin + GMS enhanced emulsion stability and texture

Final Optimised Variant (Sample H)

- Clean flavour, creamy texture, stable scoop, minimal grey hue
- Notable improvement in melt resistance and overall acceptability



KEY FINDINGS – CONSUMER ACCEPTANCE



- **Clear Preference for Sample H:** Panel consistently rated Sample H higher than Sample G across all sensory aspects.
- **Texture Breakthrough:** Sample H's texture scored 66% better, mimicking traditional ice cream creaminess.
- **Strong Overall Acceptance:** Sample H achieved a 31.7% higher overall liking, showing clear consumer favoritism.

CONCLUSION

High-quality formulation

Viable dessert base

Nutritional profile

Consumer acceptance

Lactose-free

Plant-based innovation

Allergen-friendly

Sustainable



FUTURE WORKS

Natural stabilizers

Proximate analysis

Oxidative stability

Scale-up optimization

Microbiological shelf-life

Packaging effects

Clean-label focus

Natural emulsifiers

Commercial feasibility

Micronutrient profiling

**THANK
YOU**